IIITH PGEE 2026
Memory Based Questions

General Aptitude (40 Questions)

Q1: if A, B and C complete a task individually in 12, 16, and 24 days and if A leaves after 2 days how many more days will be required to complete the entire task?
Answer: 6 more days

Q2-9: Nararyan Murthy comprehension - 4Qs and Survey comprehension 4Qs

Q10) Graph based ans was 6

11 and 12 graph based)

Q13: Two circles intersect at 2 points and a chord is drawn at 2 points of length 16. Distance between their centres is given as 21 and find radius of smaller circle if radius of larger circle is 17

Q14: total electricity consumption question. January 350 units consumed and February 80 units consumed calculate for 2 months
month 1- 350 units
month 2- 80 units
For 1-100 units charge is = 4 /unit
Similarly for 101-300= 6/ unit
And 300+ = 9/unit
fixed meter charge = 50
10% govt tax charge in overall cost
find total cost

Q15: 3 men and 3 tigers question, boat moves from one bank (all men + tiger) to another bank (no one) by taking 1 or 2 at a time, if lions outnumbered men then men will be killed. Min turns of boat so that entire journey is safe?

Q16: Three dice are rolled independently. What is probability of a case where difference between highest and lowest value on the dice is exactly 4?

Q17: What was the day on 25th April 2026?

Q18
Consider the sequence formed by writing natural numbers starting from 1 with the following rules:
Replace every multiple of 3 with “dhoni”
Replace every multiple of 5 with “kohli”
Replace every multiple of both 3 and 5 with “dhoni kohli”
Only actual numbers are counted; “dhoni”, “kohli”, and “dhoni kohli” are not counted as numbers
Example sequence (first few terms)
1, 2, dhoni, 4, kohli, dhoni, 7, 8, dhoni, kohli, 11, dhoni, 13, 14, dhoni-kohli, 16, …
What is the 100th number in this filtered sequence ?

Q19. A train running at 60kmph crosses a static pole on platform in 30 seconds. What is length of the train?

Q20. A and B compete in a 400m race with speeds 8m/s and 10m/s. A gets a headstart of 60 meters. What will be the result of the race?

Q21: : How many integers between 1 and 1000 are divisible by 3 or 7

Q23 How many positive integer factors of N = 2^2 * 3^3 * 4^4 * 5^5 ?

Q24:**: 4 statements given find out who copied the assignment if exactly one person copied and one person lied . there are 4 students . only one statement is false.**

Rohan - …
Karan* - …
Neha - rohan has copied
Aditi - neha is lying

Q25: differentiation of sinx + e^x

Q26: Minimum number of people if 2 of them are selected then the the probability that their birth month is same is >=50% ?

Q27: if 66 people like to travel by air, 128 by train and 55 via car and 15 like all three while 45 like at least two ways, find the number of people that like at least one of the ways.

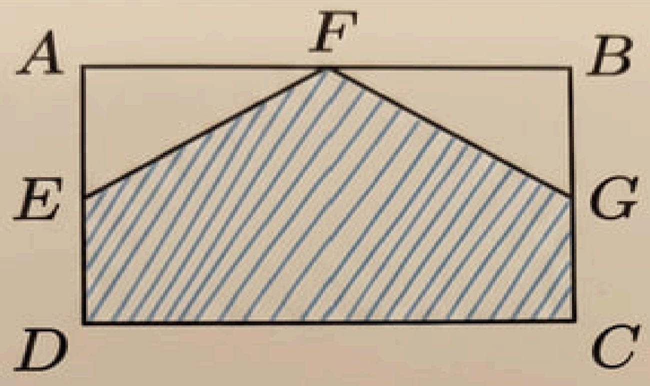
Q28: 3*3*3 cube has 2 opp sides painted red
And remaining 4 are coloured blue
How many blocks will be only blue.

Q29: 3 positive number are in arithmetic progression whose sum is 21. When X is added to smallest, 3 is added to middle and Y is added to largest, numbers turn into a geometric progression. Find 2 smallest number (7 & 3 answer tha

Q30:**circular seating arrangement question. There are 7 people . - (answer C)**

Q31

Q31: Ratio of the shaded area in the rectangle



$$\frac{AF}{BF} = \frac{1}{1}$$

$$\frac{BG}{GC} = \frac{3}{1}$$

$$\frac{EA}{ED} = \frac{1}{3}$$

Q32: : F(x) = x^2 - 3x + 2, find x for which fof(x) = 0 (answer was 0,3, 3+root5)

Q33: Power of power set is 256 (ans was 3)

Q34: A person travels 8 km north and turned 60 degree and travelled 7km and stopped, total distance to go back to starting position and in which direction (Ans. 13 km South-west)

Q35: Cylinder with height 15cm and diameter 10cm. From the top a cone of the same diameter and 6cm depth is cut. At the bottom a hemisphere of the same diameter is added. Find total volume.

Q36. Find the missing number.(25)

Column 1	Column 2	Column 3
5	3	16
10	8	36
13	12	?

The value of following is:

$$\sqrt{12+\sqrt{12+\sqrt{12+\ldots\ldots}}}$$

(a) $2\sqrt{2}$ (b) $2\sqrt{3}$

(c) 4 (d) 2

Q 37:

Q38: : Domain of fog , $f(x)=1/x-5$ and $g(x)= (x-2)^{1/2}$

Q 39: Find all combinations of 3 positive integers that add upto 10. Consider (2, 2, 6) and (6, 2, 2) as different combination.

Q40: in which of the following mean is significantly lesser than mode. (retirement ages of most people 65 and some 40)

Computer Science (40 Questions)

Q1

```
#include <stdio.h>

int main() {
   static int var = 5; printf("%d ", var--);
   if (var) {
      main();
   }
}
```

output? (Ans : 5 4 3 2 1)

Q2: Time Complexity for finding number of connected components (ans Big O(V + E))

Q3: TC for inserting n^2 elements in AVL tree:- O(n^2logn)

Q4: which Data structure will do following operations in O(n) time

1. Deleting max element
2. Searching for an element
   (answer is BST) Balanced Binary Search Tree

Q5: GCD time complexity (logn)

Q6: Which algorithm detects cycles in a Undirected graph?
a) Topological Sort b) Prims **c) Union Find** d) Dijkstras

Q7: what is the condition for validity of sql operation P-R if P and R are two database tables

Q8: Given two max Heap algorithms: 1st Program :Bottom up (using Heapify at each index starting at n/2 to 1, 2nd Program: Top down (using repeated insertion, from n = 2 to n).

Time complexity of both algorithms is O(n) Both algorithms builds the heap but gives different output Both algorithms builds the heap and give the same output.

Q9: Round Robin question with TQ = 2, all process have 0 waiting time. p1 = 4ms, p2 = 5ms, p3 = 2ms. Find wait time of p3. (ans is 4ms)

Q10: for process x, y, z which one is deadlock free? **X:**
Wait(a)
Wait(b)
Wait(c)

- **Y:**
  Wait(b)
  Wait(c)
  Wait(d)
- **Z:**
  Wait(c)
  Wait(d)
  Wait(a)

(GATE PYQ)

Q11: In client server arch with SYN, ACK, and sequence. If servers responds to client the value of Ack will be? (1+sequence, equal to sequence no, etc) (answe seq+1)

Q12: Post-order = {8, 6, 7, 3, 4, 2, 5, 1} In-order = {8, 6, 3, 7, 2, 4, 1, 5} Heigh of tree from root to leaves (Maximum path length from root to leaf) (Ans is 4 if root is considered 0)

Q13. Time complexity of creating a sorted linked list with n elements. Initially the list is empty.

Q14 Stooge sort algorithm question (3T(2N/3) + O(1)?)

Q15) number of binary trees possible with 3 unlabeled nodes

Q16) which will have most number of tuples as response - cartesian product, outer join, inner join

Q17) SQL Query Name of employee whose income is >= average income of their department.
employee(EID, NAME, DID)
Department(DID, Dname)

Q 18) If there are 100 process and 32 bit virtual address with page size 2kB, and each entry in page table is 4 bytes find the size required for all page tables

Q 19) Which of the following operation in max heap takes O(n) time,

A) Build Heap
B) N Heapify max
C) nHeap insertion

Q 20) Two threads A and B
A: X- -
B: X + +
X is shared variable and X++ , X– is not atomic
What are the possible values of X after execution X initial value is 0

Q21) Subnetting que

Que 22) A has 7 rows, B has 5 rows, 3 rows in common. How many rows does full outer join return?

Q23) Page Faults for 1, 2, 3, 4, 2, 1, 2..., 3 frame in main memory

Q24) B+ tree question
Block size is 1024
For internal node find how many keys will be required

Q25) Jacabson algo

Q26) Count Inversion question (insertion sort)

Q27) 10 threads writing data on 1000 data on a shared database (i.e., 10000 data are written).

Q 28) Amount of ROM required for 2 8 bit number multiplication

29) if m is number if input lines for a decorer and n is output what is m+n for 1KB

30) mutex lock condition que

31) Is P(S(R)) = S(P(R)) valid where s means selection and p means projection?
(selection condition uses

32) Zero has two representations in

A. Sign-magnitude
B. 1 complement
C. 2 complement
D. None of the above

33 what will be IEEE precision format for 0.5 (0,126,0)

Q34: 2s complement of 8*P. P is a 16 bit number
https://gateoverflow.in/2179/gate-cse-2010-question-8?show=350304

Q35: Which is Heap
https://gateoverflow.in/1341/gate-cse-2009-question-59

36) Given values i=-1,j=-1,k=1,l=2
M=i++&&j++llk++&&l++

37) recursion based . getchar putchar question answer was (EDCBA - reverse order of everything before space)

38) inner join 2 tuple (ans)

39) Round robin with large time quantum (FCFs ans)

40) On doint 2's complement which of the following doesn't change?
A) MSB is 1 and others 0
B)) LSB is 1 and others 0
C) Equal 0 and 1
D) Dont remember

Engineering Mathematics (10 Questions)

1. P, Q, R matrices with different dimensions. Which one of QxRxP, PxQxR, RxQxP exists
2. P(A) = 0.2, P(B) = 0.3 and P(A U B) = 0.5, then A and B are?
    - Mutually exclusive
    - Independent
    - Both a and b
    - none
3. P(A) = 0.2, P(B) = 0.3 and P(A U B) = 0.6 then P(A intersection B) = ?
4. CNF of (P->(q&r))(q->r))   { (P implies QR)and (Q implies R))
5. Number of one-to-one onto functions from {1, 2, 3, 4} to {a, b, c, d}
6. One biased coin with head prob 0.8 and 3 unbiased coins with head prob 0.5 , find prob of head
7. **Find** rank of matrix if A^-1 exists Matrix was { 1,2,b }{ 2,4,c }{ 3,a,6} (3X3 matrix thi)
8. sum of eigenvalues of a matrix. 4 x4 matrix given .
9. Which increases faster for value of x
   2^2^x
   100000^x
   e^e^logx
   e^e^x
10. . Function f R->R, f'(x) =0 at x=x*, then what can we conclude about df/dx